

Transcendental Algebraic Geometry

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Abstract

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1.1

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 1.1.1 (The Analytification of a Scheme)

Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $\alpha : X^{\text{an}} \rightarrow X$ satisfying the following universal property: For any analytic space Y and any morphism of locally ringed spaces $f : Y \rightarrow X$, there exists a unique morphism of analytic spaces $g : Y \rightarrow X^{\text{an}}$ such that the following diagram commutes:

$$\begin{array}{ccc} & Y & \\ \exists! g \swarrow & \downarrow f & \\ X^{\text{an}} & \xrightarrow{\alpha} & X \end{array}$$

Recall that the Yoneda embedding $h : \mathbf{LRS} \rightarrow \mathbf{Set}^{\mathbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \mathbf{LRS} \rightarrow \mathbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\mathbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\mathbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \rightarrow X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X , and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 1.1.2

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 1.1.3 (The Analytification Functor)

Define the analytification functor

$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \rightarrow \mathbf{An}$$

as follows.

- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X .
- For a morphism of schemes $f : X \rightarrow Y$ over $\mathbb{C}$, the universal property of Y^{an} induces a morphism $X \rightarrow Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \rightarrow X$ gives the desired map $X^{\text{an}} \rightarrow Y^{\text{an}}$.

Proposition 1.1.4

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.

Proposition 1.1.5

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the following are true.

- If X is separated, then X^{an} is Hausdorff.
- If X is proper, then X^{an} is proper.

Proposition 1.1.6

Let X be a variety over $\mathbb{C}$. Then the following are true.

- X is complete if and only if X^{an} is compact.
- If X is smooth, then X^{an} is a complex manifold.
- If X is irreducible, then X^{an} is connected.

Theorem 1.1.7

Let X be a smooth variety over $\mathbb{C}$. Then there is a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{dR}}^n(X^{\mathrm{an}})$$

induced by the canonical map $\alpha : X^{\mathrm{an}} \rightarrow X$.

Corollary 1.1.8

Let X be a smooth variety over $\mathbb{C}$. Then the above isomorphism and de Rham's theorem induces a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{sing}}^n(X^{\mathrm{an}}; \mathbb{C})$$

Theorem 1.1.9 (The Period Isomorphism)

Let k be a field. Let X be a smooth variety over k . Then there is a natural isomorphism

$$H_{\mathrm{dR}}^*(X) \otimes_k \mathbb{C} \cong H_{\mathrm{sing}}^*(X^{\mathrm{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

1.2**Proposition 1.2.1**

Let X be a smooth projective curve over $\mathbb{C}$. Then X^{an} is homeomorphic to $\Sigma_{p_g(X)}$ the orientable surface of genus $p_g(X)$.